Phosphocholine Antagonizes Listeriolysin O-Induced Host Cell Responses of Listeria monocytogenes.

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Bacterial toxins disrupt plasma membrane integrity with multitudinous effects on host cells. The secreted pore-forming toxin listeriolysin O (LLO) of the intracellular pathogen Listeria monocytogenes promotes egress of the bacteria from vacuolar compartments into the host cytosol often without overt destruction of the infected cell. Intracellular LLO activity is tightly controlled by host factors including compartmental pH, redox, proteolytic, and proteostatic factors, and inhibited by cholesterol. Combining infection studies of L. monocytogenes wild type and isogenic mutants together with biochemical studies with purified phospholipases, we investigate the effect of their enzymatic activities on LLO. Here, we show that phosphocholine (ChoP), a reaction product of the phosphatidylcholine-specific phospholipase C (PC-PLC) of L. monocytogenes, is a potent inhibitor of intra- and extracellular LLO activities. Binding of ChoP to LLO is redox-independent and leads to the inhibition of LLO-dependent induction of calcium flux, mitochondrial damage, and apoptosis. ChoP also inhibits the hemolytic activities of the related cholesterol-dependent cytolysins (CDC), pneumolysin and streptolysin. Our study uncovers a strategy used by L. monocytogenes to modulate cytotoxic LLO activity through the enzymatic activity of its PC-PLC. This mechanism appears to be widespread and also used by other CDC pore-forming toxin-producing bacteria.

Listeriolysin O: a phagosome-specific lysin.

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Listeriolysin O (LLO) is a pore-forming toxin of the cholesterol-dependent cytolysin family and a primary virulence factor of the gram-positive, facultative intracellular pathogen Listeria monocytogenes. During the intracellular life cycle of L. monocytogenes, LLO is largely responsible for mediating rupture of the phagosomal membrane, thereby allowing the bacterium access to the host cytosol, its replicative niche. In the host cytosol, LLO activity is controlled at numerous levels to prevent perforation of the plasma membrane and loss of the intracellular environment. In this review, we focus primarily on the role of LLO in phagosomal escape and the multiple regulatory mechanisms that control LLO activity in the host cytosol.

Listeriolysin is processed efficiently into an MHC class I-associated epitope in Listeria monocytogenes-infected cells.

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Listeria monocytogenes is an intracellular pathogen that enters the cytoplasm of infected cells by secreting listeriolysin (LLO), a protein that destroys the phagosomal membrane. In infected mice, LLO is a major Ag detected by protective, MHC class I-restricted CTLs. Although the role of LLO in pathogenesis and host immunity is well established, its rate of intracellular synthesis has yet to be determined. Herein we show that cytosolic L. monocytogenes secrete LLO at a relatively low rate of approximately one molecule per bacterium per minute. Under extracellular labeling conditions, the rate of LLO secretion is approximately 50-fold higher. Intracellular LLO synthesis suffices, however, for the accumulation of 600 to 1000 H-2Kd-associated LLO 91-99 epitopes per cell. We calculate that between four and 11 LLO molecules are degraded for each LLO 91-99 epitope bound by H-2Kd. Our findings indicate that the antigenicity of LLO, with respect to MHC class I-restricted CTLs, cannot be attributed to high levels of intracellular secretion. Rather, LLO is a dominant Ag because it is rapidly degraded and very efficiently processed into an MHC class I-associated epitope.

The cell biology of Listeria monocytogenes infection: the intersection of bacterial pathogenesis and cell-mediated immunity.

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Listeria monocytogenes has emerged as a remarkably tractable pathogen to dissect basic aspects of cell biology, intracellular pathogenesis, and innate and acquired immunity. In order to maintain its intracellular lifestyle, L. monocytogenes has evolved a number of mechanisms to exploit host processes to grow and spread cell to cell without damaging the host cell. The pore-forming protein listeriolysin O mediates escape from host vacuoles and utilizes multiple fail-safe mechanisms to avoid causing toxicity to infected cells. Once in the cytosol, the L. monocytogenes ActA protein recruits host cell Arp2/3 complexes and enabled/vasodilator-stimulated phosphoprotein family members to mediate efficient actin-based motility, thereby propelling the bacteria into neighboring cells. Alteration in any of these processes dramatically reduces the ability of the bacteria to establish a productive infection in vivo.

The Listeriolysin O PEST-like Sequence Co-opts AP-2-Mediated Endocytosis to Prevent Plasma Membrane Damage during Listeria Infection.

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Listeriolysin O (LLO) is a cholesterol-dependent cytolysin that mediates escape of Listeria monocytogenes from a phagosome, enabling growth of the bacteria in the host cell cytosol. LLO contains a PEST-like sequence that prevents it from killing infected cells, but the mechanism involved is unknown. We found that the LLO PEST-like sequence was necessary to mediate removal of LLO from the interior face of the plasma membrane, where it coalesces into discrete puncta. LLO interacts with Ap2a2, an adaptor protein involved in endocytosis, via its PEST-like sequence, and Ap2a2-dependent endocytosis is required to prevent LLO-induced cytotoxicity. An unrelated PEST-like sequence from a human G protein-coupled receptor (GPCR), which also interacts with Ap2a2, could functionally complement the PEST-like sequence in L. monocytogenes LLO. These data revealed that LLO co-opts the host endocytosis machinery to protect the integrity of the host plasma membrane during L. monocytogenes infection.

Molecular and cellular basis of the infection by Listeria monocytogenes: an overview.

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This review rather than covering the whole field, intends to highlight the particularly interesting properties of some proteins involved in the infection by Listeria monocytogenes and the general interest in some of the approaches used to analyze the molecular and cellular basis of the infection. After an introduction to the bacterium and to the disease, a description of the infection at the cellular level will be given. The specific features of the pore-forming toxin listeriolysin O, as a protein particularly well adapted to the intracellular lifestyle of L. monocytogenes will be discussed. By describing in detail how the bacterium moves inside cells, particular attention will be given to show how addressing this issue has provided key answers and instrumental tools to cell biologists studying actin-based motility. The analysis of the entry process and in particular the studies derived from the specificity of internalin for its receptor will demonstrate how an apparently reductionist approach can help generating relevant animal models, identification of virulence factors and demonstration of their role in vivo. The virulence gene cluster and its regulation by PrfA will be presented and discussed in the framework of the recently determined genome sequence.

Listeriolysin S Is a Streptolysin S-Like Virulence Factor That Targets Exclusively Prokaryotic Cells In Vivo.

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Streptolysin S (SLS)-like virulence factors from clinically relevant Gram-positive pathogens have been proposed to behave as potent cytotoxins, playing key roles in tissue infection. Listeriolysin S (LLS) is an SLS-like hemolysin/bacteriocin present among Listeria monocytogenes strains responsible for human listeriosis outbreaks. As LLS cytotoxic activity has been associated with virulence, we investigated the LLS-specific contribution to host tissue infection. Surprisingly, we first show that LLS causes only weak red blood cell (RBC) hemolysis in vitro and neither confers resistance to phagocytic killing nor favors survival of L. monocytogenes within the blood cells or in the extracellular space (in the plasma). We reveal that LLS does not elicit specific immune responses, is not cytotoxic for eukaryotic cells, and does not impact cell infection by L. monocytogenes Using in vitro cell infection systems and a murine intravenous infection model, we actually demonstrate that LLS expression is undetectable during infection of cells and murine inner organs. Importantly, upon intravenous animal inoculation, L. monocytogenes is found in the gastrointestinal system, and only in this environment LLS expression is detected in vivo Finally, we confirm that LLS production is associated with destruction of target bacteria. Our results demonstrate therefore that LLS does not contribute to L. monocytogenes tissue injury and virulence in inner host organs as previously reported. Moreover, we describe that LlsB, a putative posttranslational modification enzyme encoded in the LLS operon, is necessary for murine inner organ colonization. Overall, we demonstrate that LLS is the first SLS-like virulence factor targeting exclusively prokaryotic cells during in vivo infections.IMPORTANCE The most severe human listeriosis outbreaks are caused by L. monocytogenes strains harboring listeriolysin S (LLS), previously described as a cytotoxin that plays a critical role in host inner tissue infection. Cytotoxic activities have been proposed as a general mode of action for streptolysin S (SLS)-like toxins, including clostridiolysin S and LLS. We now challenge this dogma by demonstrating that LLS does not contribute to virulence in vivo once the intestinal barrier has been crossed. Importantly, we show that intravenous L. monocytogenes inoculation leads to bacterial translocation to the gastrointestinal system, where LLS is specifically expressed, targeting the host gut microbiota. Our study highlights the heterogeneous modes of action of SLS-like toxins, and we demonstrate for the first time a further level of complexity for SLS-like biosynthetic clusters as we reveal that the putative posttranslational modification enzyme LlsB is actually required for inner organ colonization, independently of the LLS activity.

Acacetin Alleviates Listeria monocytogenes Virulence Both In Vitro and In Vivo via the Inhibition of Listeriolysin O.

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Listeria monocytogenes is a ubiquitous Gram-positive foodborne pathogen that is responsible for listeriosis in both humans and several animal species. The bacterium secretes a pore-forming cholesterol-dependent cytolysin, listeriolysin O (LLO), a major virulence factor involved in the activation of cellular processes. The ability of LLO to lyse erythrocytes is a measure of LLO activity. We used hemolytic activity assay to screen the LLO inhibitors. Acacetin was found to be an LLO inhibitor, which is a di-hydroxy and mono-methoxy flavone present in various plants, including Black locust, Damiana, and Silver birch. As the features of acacetin are of low toxicity and have less acquired resistance, it comes to a hotspot in drug development. In our study, we report that acacetin antagonized the hemolytic activity of L. monocytogenes culture supernatants and purified LLO by directly interfering with the formation of oligomers without inhibiting the bacterial growth and the expression of LLO. Acacetin also relieved the injury of alveolar epithelial cells by inhibiting LLO activity. Further, acacetin significantly promoted the clearance of L. monocytogenes and alleviated the histopathological damage, thereby raising survival rate, which conferred mice with effective protection against L. monocytogenes infection. Using molecular docking and dynamics simulation, we further proved the mechanism of acacetin antagonizing LLO pore-forming activity by direct binding to the second membrane-inserting helix bundle (HB2) of LLO domain 3. These data suggested that acacetin recedes the virulence of L. monocytogenes both in vivo and in vitro, and this study provided a promising candidate and potential alternative for the prevention and treatment of L. monocytogenes infections.

The molecular mechanisms of listeriolysin O-induced lipid membrane damage.

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Listeria monocytogenes is an intracellular food-borne pathogen that causes listeriosis, a severe and potentially life-threatening disease. Listeria uses a number of virulence factors to proliferate and spread to various cells and tissues. In this process, three bacterial virulence factors, the pore-forming protein listeriolysin O and phospholipases PlcA and PlcB, play a crucial role. Listeriolysin O belongs to a family of cholesterol-dependent cytolysins that are mostly expressed by gram-positive bacteria. Its unique structural features in an otherwise conserved three-dimensional fold, such as the acidic triad and proline-glutamate-serine-threonine-like sequence, enable the regulation of its intracellular activity as well as distinct extracellular functions. The stability of listeriolysin O is pH- and temperature-dependent, and this provides another layer of control of its activity in cells. Moreover, many recent studies have demonstrated a unique mechanism of pore formation by listeriolysin O, i.e., the formation of arc-shaped oligomers that can subsequently fuse to form membrane defects of various shapes and sizes. During listerial invasion of host cells, these membrane defects can disrupt phagosome membranes, allowing bacteria to escape into the cytosol and rapidly multiply. The activity of listeriolysin O is profoundly dependent on the amount and accessibility of cholesterol in the lipid membrane, which can be modulated by the phospholipase PlcB. All these prominent features of listeriolysin O play a role during different stages of the L. monocytogenes life cycle by promoting the proliferation of the pathogen while mitigating excessive damage to its replicative niche in the cytosol of the host cell.

Listeria monocytogenes exploits efferocytosis to promote cell-to-cell spread.

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Efferocytosis, the process by which dying or dead cells are removed by phagocytosis, has an important role in development, tissue homeostasis and innate immunity. Efferocytosis is mediated, in part, by receptors that bind to exofacial phosphatidylserine (PS) on cells or cellular debris after loss of plasma membrane asymmetry. Here we show that a bacterial pathogen, Listeria monocytogenes, can exploit efferocytosis to promote cell-to-cell spread during infection. These bacteria can escape the phagosome in host cells by using the pore-forming toxin listeriolysin O (LLO) and two phospholipase C enzymes. Expression of the cell surface protein ActA allows L. monocytogenes to activate host actin regulatory factors and undergo actin-based motility in the cytosol, eventually leading to formation of actin-rich protrusions at the cell surface. Here we show that protrusion formation is associated with plasma membrane damage due to LLO's pore-forming activity. LLO also promotes the release of bacteria-containing protrusions from the host cell, generating membrane-derived vesicles with exofacial PS. The PS-binding receptor TIM-4 (encoded by the Timd4 gene) contributes to efficient cell-to-cell spread by L. monocytogenes in macrophages in vitro and growth of these bacteria is impaired in Timd4(-/-) mice. Thus, L. monocytogenes promotes its dissemination in a host by exploiting efferocytosis. Our results indicate that PS-targeted therapeutics may be useful in the fight against infections by L. monocytogenes and other bacteria that use similar strategies of cell-to-cell spread during infection.